

White Chang

Senior Backend Engineer

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Senior backend engineer with nearly 15 years of software development experience, specializing in high-concurrency distributed systems. I focus on solving problems — performance bottlenecks, deployment automation, system security — not memorizing frameworks. Domain experience spans real-time gaming platforms (multi-server architectures serving multiple concurrent titles), payment processing, and enterprise APIs. My workflow is AI-augmented: Claude Code runs as my main session for planning and final decisions, delegating file search, implementation, and verification to subagents on different models (whoever writes the code never verifies it); harness rule files keep AI behavior in check, and my self-built memory-mcp carries context across sessions. I catch what AI misses — production observability, security hardening, system integrity.

Core Competencies

System Modernization & Multi-Tenancy

Solo-led full platform rewrite from Lua/PHP/Java to Go with config-driven multi-tenant architecture, replacing 13 per-platform release branch forks with a single codebase. Designed Proto2/Proto3 dual-compatible serialization layer for seamless legacy client migration.

Golang, gRPC, Protobuf, Redis, MySQL, MongoDB

High-Concurrency Backend Architecture

Designed multi-server distributed systems (Gate/Lobby/Game) across 10+ repositories and multiple platforms. Built reusable frameworks that cut new title development cycles by over 50%.

Lua/Skynet, Golang, TCP/WebSocket, gRPC

Database & Cache Optimization

Diagnosed full-collection scans on ~2.7M records/day and designed compound indexes. Restructured Redis key architecture to eliminate excessive SCAN operations.

MongoDB, Redis, MySQL, PostgreSQL

DevOps & CI/CD

Automated Docker image builds to Harbor registry, designed Helm chart deployments on Kubernetes, set up GitLab Runners and remote SSH deployments.

Kubernetes, Docker, Helm, GitLab CI/CD

Security & Data Integrity

Investigated Redis data tampering incidents, analyzed full attack chains. Designed HMAC-SHA256 data signing mechanism to architecturally prevent direct Redis manipulation that bypasses application-layer validation.

AI-Augmented Development

Claude Code is my main session for day-to-day work — debugging, investigation, and architecture discussions happen there, with me owning planning and final decisions. Implementation and verification are delegated to subagents on different models depending on the task.

- Multi-model subagent delegation: lightweight models handle file search and routine chores, a worker handles implementation, and a separate, independent verifier signs off — whoever writes the code never verifies it
- Architect-first workflow: I make architecture decisions (module boundaries, protocols, security model), then use AI agents for implementation — catching production blind spots that AI overlooks, like logging coverage and security hardening
- Engineering guardrails for AI: harness rule files (CLAUDE.md / AGENTS.md) constrain AI behavior and enforce project conventions; my self-built memory-mcp provides cross-session memory so preferences, past pitfalls, and unfinished work carry over to the next session
- Cross-stack delivery: nearly 15 years across Golang, Java, Lua, JavaScript — combined with AI collaboration, enables rapid context-switching across any tech stack

Employment History

- **Senior Backend Engineer** · Tyche Tech Co, Ltd — Jan. 2021 — Present
  3 game server projects (Lua/Skynet, Golang). Fixed reward-claim race condition, MongoDB optimization on 2.7M records/day. Redis architecture redesign. Security incident investigation & AES encryption. CI/CD pipeline with Docker/Harbor/Helm. Handled containerization of multiple platforms, migrating from bare metal (Harbor/ECR releases, cutover runbooks, data verification scripts); introduced a shared team knowledge base and AI collaboration rules.
- **Senior Backend Engineer** · Terpro Mobile Co, Ltd — Jun. 2019 — Dec. 2020
  Built a reusable game mechanics framework, cutting new-game development time by over 50%. Kubernetes/Helm deployment.
- **Senior Backend Engineer** · WistronITS Corp. Stationed: Taiwan Mobile — Sep. 2016 — May 2019
  Led 2 engineers. Built 5 micro-services (OpenId, Product, Payment, Stock, Kafka). Jenkins/Nexus CI toolchain.
- **Software Engineer** · Kangda Info Corp. — Jul. 2015 — Aug. 2016
- **Software Engineer** · BenQ Corp. — Nov. 2013 — Feb. 2015
- **Software Engineer** · Fashion Intelligence Co, Ltd (Open-Life) — Nov. 2011 — Oct. 2013

Freelance & Side Projects

- **memory-mcp — Cross-Session Memory for AI Coding Agents (open source, personal)** — Jun. 2026 — Present
- **4x — Multi-Role AI Development Framework (open source, archived)** — Jun. 2026 — Aug. 2026
- **PGS Game Server (solo)** — Dec. 2022 — Apr. 2023
- **Yapro car tax refund management (2 members)** — Jan. 2021 — Mar. 2021
- **Muxin CarePro CMS (5 members)** — Nov. 2013 — Mar. 2015

Education

Bachelor of Electrical Engineering — National Kaohsiung University of Science and Technology(NKUST), Taiwan — Sep. 2006 — Jun 2010